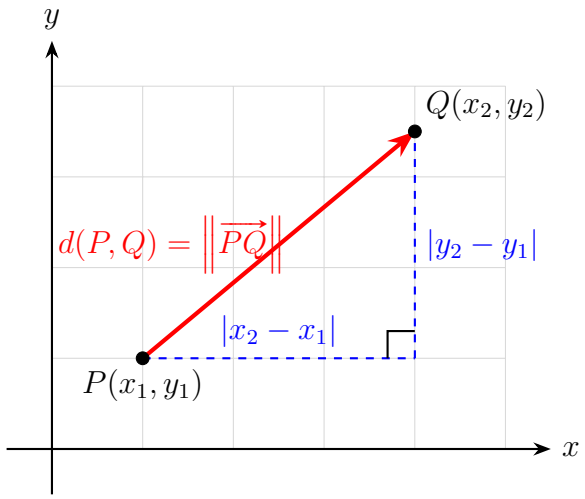
2 Discussion Week 1: Distance Formulae and Geometric Shapes

Definition 2.1: Distance Formula in $\mathbb{R}^2$

Let $(x, y) \in \mathbb{R}^2$ and points $P = (x_1, y_1)$, $Q = (x_2, y_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance formula is a direct application of the Pythagorean theorem, where the distance $d(P, Q)$ forms the hypotenuse of a right-angled triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$, as shown in the figure below.



The case in $\mathbb{R}^3$ follows similarly:

Definition 2.2: Distance Formula in $\mathbb{R}^3$

Let $(x, y, z) \in \mathbb{R}^3$ and points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2.1 Equations of Geometric Shapes

Consider the equation $x^2 + y^2 = 4$.

In $\mathbb{R}^2$, this equation is equivalent to the set:

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

This represents a circle with radius $r = 2$ centered at the origin. The distance from the origin $(0, 0)$ to any point (x, y) on this circle is constant and equal to 2, which can be verified directly using the distance formula:

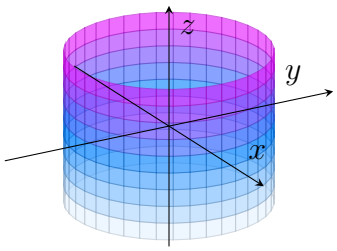
$$\sqrt{(x-0)^2 + (y-0)^2} = 2$$

Hence, the geometry of the circle is directly implied by the Pythagorean theorem.

Now consider the equation $x^2 + y^2 = 1$ in three-dimensional space, $\mathbb{R}^3$. This set is formally expressed as:

$$\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$$

Because z is completely unrestricted, the cross-section at any constant height $z = c$ forms the unit circle $x^2 + y^2 = 1$. Consequently, this equation describes a *right circular cylinder* extending infinitely along the z-axis, as shown.



2.2 Linear Equations across Dimensions

In $\mathbb{R}^2$, the general form of a linear equation is given by:

$$Ax + By + C = 0 \quad (A, B, C \in \mathbb{R}, \text{ with } A \text{ and } B \text{ not both zero})$$

This equation represents a *line* in two-dimensional space.

In $\mathbb{R}^3$, however, the exact same algebraic equation $Ax + By + C = 0$ represents a *plane* parallel to the z-axis (since z is a free variable), not a line.

To uniquely define a line in three-dimensional space, a single linear equation is insufficient; we require a system of *two* independent linear equations (representing the intersection of two non-parallel planes).